

Prognostic factors influencing event-free survival and treatments in desmoid-type fibromatosis: analysis from a large institution

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Abstract

BACKGROUND: Prognostic factors and optimal management of desmoid tumors have been discussed for decades. The authors present the results of a large series of patients with desmoid tumors treated at a single institution to investigate the prognostic factors influencing event-free survival (EFS) and suitable treatments for these rare tumors.

METHODS: Two hundred fourteen patients with desmoid tumors admitted to the surgical department were included, of whom 20 were recommended for a policy of watchful waiting. The following clinical parameters were studied: admission status, age, sex, tumor site, tumor size, margin status, and therapeutic strategy. Univariate and multivariate analysis were performed for EFS.

RESULTS: Forty-two patients had local recurrence. One patient died of intra-abdominal disease. The 5-year and 10-year EFS rates were 78.8% and 77.9%, respectively. In univariate analysis, admission status, tumor site, tumor size, and group (R0 vs R1 and R0 vs R2) had significant impacts on EFS. EFS discrepancy was not significant between R1 and R2 or biopsy groups. In multivariate analysis, tumor size and admission status had independent value. The median delay to progression for patients undergoing watchful waiting was comparable with that for the surgical group.

CONCLUSIONS: This study demonstrates that tumor size and a history of recurrence are independent predictors of EFS. Surgery is warranted if it can be R0 and function sparing. Nonsurgical modalities or a policy of watchful waiting may be a better choice for unresectable disease.

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Desmoid tumors, also called desmoid-type fibromatosis (DF), are rare soft tissue tumors that are deep-seated, musculoaponeurotic-derived monoclonal proliferations of

fibroblastic cells that locally infiltrate but do not metastasize.^{1,2} DF has a relatively low incidence of 2.4 to 4.3 per 1 million annually but a high tendency for local recurrence, even after apparently radical resection.³ It can be classified by origin into extra-abdominal, intra-abdominal, and abdominal types. It is more frequent in women than men, particularly the abdominal type. Most desmoid tumors develop sporadically in young adults, with a small intra-abdominal subgroup associated with familial adenomatous polyposis and Gardner's syndrome. Although the exact etiology of

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desmoid tumors is unknown, genetic abnormalities, trauma, and steroid sex hormones may contribute to their oncogenesis.

For decades, the standard treatment for DF has been wide local excision with adjuvant radiotherapy selectively, which serves as the optimal method to achieve eradication of the disease. However, local 5-year recurrence rates have ranged from 20% to 60% in reported retrospective studies.^{4–8} The great variety of recurrence rates may be ascribed to differences in accrual, treatment, follow-up, and intrinsic natural processes of this rare disease. Because of the ambiguous prognostic significance of positive excision margins and the absence of metastatic potential in DF, function-preserving surgery has been advocated, with particular emphasis on limiting unnecessary morbidity.^{8,9} Patients with long-term growth arrest and spontaneous regression have been reported in the literature.^{10,11}

A policy of watchful waiting has been advocated in a subgroup of patients with stable primary lesions or unresectable recurrent disease. Furthermore, desmoid tumors may respond to radiation, chemotherapy, or other systemic therapy, including molecular targeted therapy.^{12–15} Thus, a multidisciplinary approach has been increasingly recommended at large referral centers. In this case, it is of greater importance to identify patients who may benefit from surgery to achieve higher curability. The purposes of this study were to investigate prognostic factors for event-free survival (EFS) and identify suitable treatments for this rare disease.

Methods

Patient selection

All patients with diagnoses of DF who were referred to the surgical department between March 1987 and December 2009 at Fudan University Shanghai Cancer Center were retrospectively analyzed. The dates of initial treatment, recurrence or progression, and latest follow-up were recorded. The data reviewed included admission status, sex, age at presentation, tumor site, tumor size, margin status, and therapeutic strategies. Desmoid histology was confirmed for all patients by an experienced pathologist. Written informed consent was obtained from all patients. This study was approved by the ethics committee of Fudan University Shanghai Cancer Center. The study was retrospective.

Grouping and therapeutic strategies

Resection margins of patients were carefully identified by a pathologist. Patients were retrospectively categorized into 3 groups for analysis according to resection thoroughness and margin status: group I (patients underwent complete tumor resection with negative margins [R0]), group II (patients underwent complete resection but with

microscopic positive margin [R1]), and group III (patients either underwent incomplete resection with macroscopic residual disease [R2] or underwent biopsy alone for unresectable disease). During this period, the treatment policy was to operate on all patients for whom resection was deemed feasible.

Other therapeutic approaches, consisting of radiation, tamoxifen therapy, and chemotherapy, were administered to patients who had unresectable disease, underwent incomplete resection, or were at high predicted risk for recurrence on the basis of narrow margins, large tumor size, or multifocal disease. Radiation therapy was delivered by electron beam to selected patients at doses of 45 to 75 Gy on the basis of tumor load, site, and patient tolerance, at the discretion of the operating surgeon and radiologists. Systemic therapy was most commonly tamoxifen, prescribed by the operating surgeon at a daily dose of 30 mg for selected patients deemed at high risk for recurrence, while doxorubicin-based chemotherapy was given to 1 patient in the neoadjuvant setting. A subgroup of patients with unresectable disease who were unable to tolerate chemotherapy or radiotherapy were referred for observation alone.

Watchful waiting

A nonaggressive approach was adopted in a subgroup of patients in an attempt to avoid aggressive surgery. Nine patients received initial medical surveillance after surgical biopsy for unresectable disease. Eleven patients with recurrent unresectable tumors were also recommended for observation alone. Patients receiving this watchful waiting policy were mainly those with unresectable tumors or with progressive disease that could not be life threatening or cause functional loss.

Follow-up

Recorded events were recurrence and progression. Tumor response was evaluated using computed tomography every 3 months after surgery. Recurrence was defined as the appearance of a new lesion after macroscopically complete resection. Progression was defined as a 25% increase in the largest diameter of residual disease or the appearance of a new lesion after macroscopically incomplete surgery. After nonsurgical treatment, progression was defined as the appearance of a new lesion or a 25% increase in the largest diameter of the target lesion.¹⁶

Statistical analysis

EFS curves were calculated using the Kaplan-Meier method, and comparisons between curves were made using the log-rank test. Age, sex, tumor size, tumor location, admission status, group, and adjuvant therapies were tested by univariate analysis as possible prognostic factors. Parameters identified as significantly correlated with

outcomes were further tested by multivariate analysis using the Cox proportional-hazards model. All reported *P* values are 2 sided. *P* values < .05 were considered statistically significant. Median follow-up duration was 55 months, over 22 years of accrual. Five-year and 10-year estimated EFS rates were chosen as the most suitable measures of outcomes on the basis of our data.

Results

Clinical characteristics and treatment

From March 1, 1987, to December 31, 2009, 214 consecutive patients with desmoid tumors presented at the surgical department of Fudan University Shanghai Cancer Center. Table 1 shows characteristics and treatment factors for these patients. The median age was 33 years (range, 7 to 86 years). The ratio of male to female patients was approximately 1:3.4. Tumors were located extra-abdominally in

201 patients and intra-abdominally in 13 patients. Tumor size ranged from 1.5 to 35 cm. The median size of primary tumors was 5.5 cm, and the median size of recurrent tumors was 8 cm.

One hundred fifty-three patients (71.5%) presented with masses, 23 of whom also had associated pain. In these cases, however, the pain was more of an ache or discomfort only. Twenty-six patients (12.1%) presented with burning pain or numbness in the affected area. Thirty-five patients (16.4%) had nonlocalized discomfort. The duration of mass or symptoms before presentation ranged from 2.0 to 36 months, with a mean duration of 8 months. Thirty-seven female patients had histories of cesarean section before presentation, and 17 patients had histories of injury.

Surgery was recommended for all resectable tumors and was combined with radiation or tamoxifen therapy if there was evidence of positive resection margin or residual disease. Of the 214 patients enrolled, group I contained 142 patients who underwent R0 resection, including 79 (55.6%) with abdominal wall tumors, of whom 32 received

Table 1 Clinical parameters and predictors of EFS

Factor	No	EFS rate		Log-rank <i>P</i> (univariate)	Cox significance (multivariate)
		5-year	10-year		
Treatment period				.905	
Before 2000	87	79.2	77.8		
After 2000	127	77.8	77.8		
Admitted status				.000	.012
Primary	153	84.9	84.9		
Recurrent	61	63.0	58.2		
Tumor size (cm)				.000	.006
<5	63	91.1	91.1		
5–10	99	80.6	72.7		
>10	50	59.1	54.2		
Group				.003	.096
I (R0)	142	85.5	84.3		
II (R1)	35	63.3	63.3		
III (R2/biopsy)	37	63.9	63.9		
Age (y)				.659	
≤30	89	80	80		
>30	125	79	76.3		
Sex				.057	
Female	165	82.4	80.5		
Male	49	66.9	66.9		
Site				.001	.745
Extremity/girdles	51	61.0	61.0		
Head/neck	36	69.2	69.2		
Abdominal wall	91	92.5	90.7		
Thorax/back	23	79.1	79.1		
Intra-abdominal	13	69.2	69.2		
Adjuvant radiation				.062	
No	156	83.1	81.9		
Yes	58	63.7	63.7		
Tamoxifen therapy				.074	
No	180	81.9	80.8		
Yes	34	64.7	64.7		

EFS = event-free survival.

prosthetic implant reconstruction. Group II contained 35 patients who underwent R1 resection, including 10 (28.6%) with abdominal wall tumors, while group III contained 37 patients who underwent R2 resection or biopsy alone, including 2 (5.4%) with abdominal wall tumors. No patients underwent amputation.

Adjuvant therapies were administered according to the rationale explained above. The median dose for all patients who received radiotherapy was 50 Gy. In group I, 35 patients had large tumors and marginal resection. Among them, 15 patients received adjuvant radiotherapy and 20 were administered tamoxifen. One patient received doxorubicin-based chemotherapy for multifocal disease before surgery. In group II, 10 patients received postoperative radiotherapy, and 9 patients had tamoxifen therapy.

There were 37 patients in group III, including 26 who underwent incomplete resection and 9 who underwent biopsy alone. Among the incompletely resected patients, 22 received radiotherapy, and 4 received tamoxifen therapy. Nine biopsy patients did not receive further treatment and were only observed until disease progression.

Outcomes

One patient died during follow-up because of an uncontrolled intra-abdominal desmoid tumor. The median follow-up interval was 55.5 months (range, 3 to 280 months). The estimated 10-year overall survival rate was 99.5%. Estimated 5-year and 10-year EFS rates were 78.8% (95% confidence interval [CI], 75.7% to 81.9%) and 77.9% (95% CI, 74.7% to 81.1%), respectively. In total, 42 patients developed local recurrence. Twenty-two recurrences were in patients with primary tumors (14.4% [22 of 153]), whereas 20 were in patients presenting with recurrent tumors (32.8% [20 of 61]). Local recurrences were located in the extremities (29.4% [15 of 51]), abdominal wall (8.8% [8 of 91]), head and neck (30.6% [11 of 36]), thorax and back (17.4% [4 of 23]), and viscera (30.8% [4 of 13]). Local recurrences occurred in group I (14.1% [20 of 142]),

group II (34.3% [12 of 35]), and group III (27.0% [10 of 37]), including 3 recurrences in the 9 patients who underwent biopsy only.

Time to relapse ranged from 4 to 167 months (median, 16 months); 3 patients relapsed after >60 months. Univariate analysis identified admission status, tumor site, tumor size, and group (R0, R1, or R2 or biopsy) as significantly correlated with outcomes, while tumor size and admission status were independent prognostic factors based on multivariate analysis.

Patients with primary disease had relatively better outcomes, with a 5-year EFS rate of 84.9% (95% CI, 78.6% to 91.2%), unchanged at 10 years, but patients referred with recurrent disease had a 5-year EFS rate of 63.0% (95% CI, 49.5% to 76.5%) and a 10-year EFS rate of 58.2% (95% CI, 42.7% to 73.7%) ($P = .000$).

Patients with tumors <5 cm in diameter had significantly better outcomes, with a 5-year EFS of 91.1% (95% CI, 83.7% to 98.5%), unchanged at 10 years, while patients with tumors between 5 and 10 cm in diameter had a 5-year EFS of 79.1% (95% CI, 70.5% to 87.7%), unchanged at 10 years, whereas patients with tumors >10 cm in diameter had a 5-year EFS rate of 59.1% (95% CI, 43.2% to 75%) and a 10-year EFS rate of 54.2% (95% CI, 37% to 71.4%) ($P = .000$; Fig. 1).

Patients with tumors in the abdominal wall and chest wall or back had better outcomes, with 5-year EFS rates of 92% and 79.1%, respectively. However, patients with head and neck or intra-abdominal tumors had similarly worse outcomes, with 5-year EFS rate of 69.2%. Extremity tumors had the worst outcomes, with a 5-year EFS of 61%. The survival difference between each tumor site was statistically significant ($P = .001$).

Group comparison

Patients in group I had significantly better outcomes, with a 5-year EFS rate of 85.5% (95% CI, 79.2% to 91.8%) and a 10-year EFS rate of 84.3% (95% CI, 77.6% to 91%),

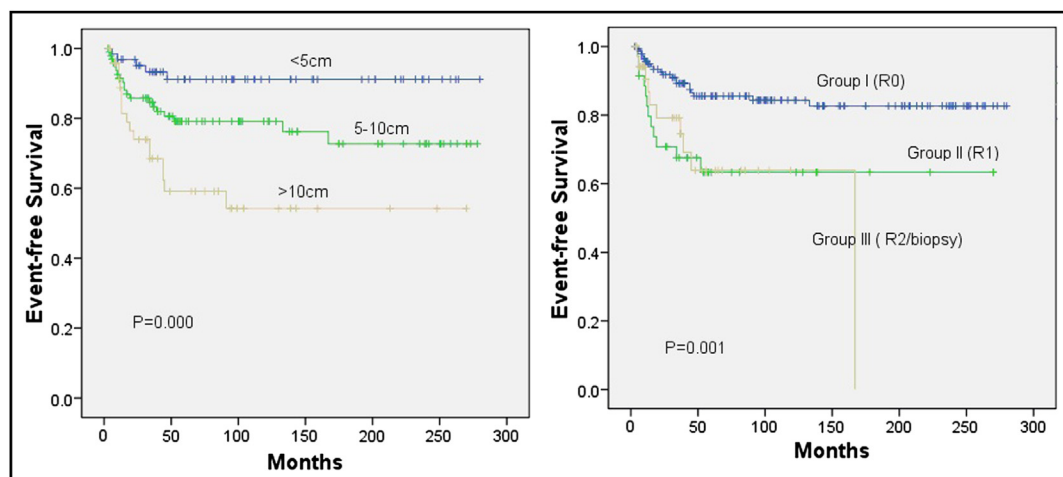


Figure 1 EFS by size and group (n = 214). Tick marks overlapping survival curves denote censored times.

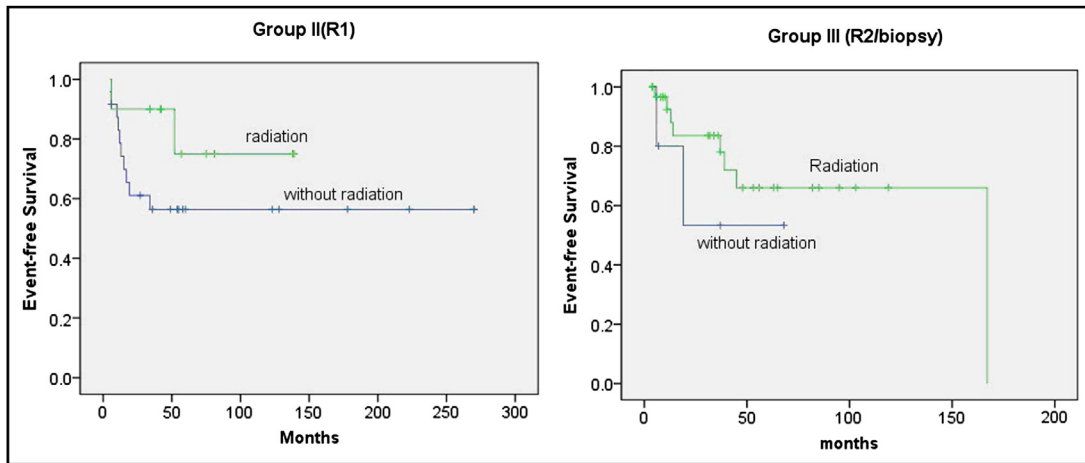


Figure 2 Effect of radiation on EFS in subgroups of groups II and III.

while patients in group II had a 5-year EFS rate of 63.3% (95% CI, 46.5% to 80.1%), unchanged at 10 years, and patients in group III had a 5-year EFS rate of 63.9% (95% CI, 44.1% to 83.7%), unchanged at 10 years. The survival differences between groups I and II and between groups I and III were statistically significant ($P = .002$ and $P = .003$), but there was not a significant survival difference between groups II and III ($P = .978$) (Fig. 1).

Radiation therapy was not shown to have a significant effect on outcomes. However, in subgroup analysis, group II patients who received adjuvant radiation had relatively better outcomes, with a 5-year EFS rate of 75%, compared with 56.4% in group II patients without adjuvant radiation ($P = .21$). Patients in group III who received radiation therapy did not have apparently better outcomes, with a 5-year EFS rate of 66%, compared with 53.3% in patients without radiation therapy ($P = .34$) (Fig. 2).

Overall, 42 recurrences were initially treated either by surgery alone ($n = 10$), surgery combined with radiation therapy ($n = 8$), radiation alone ($n = 9$), tamoxifen therapy ($n = 4$), or observation ($n = 11$). Among those patients

treated with procedures other than observation, 12 patients were free of disease, including 1 who had complete tumor regression after radiation therapy, and 11 patients had stable disease during the follow-up interval. Eight patients had second recurrences and were treated with surgery combined with additional therapies ($n = 3$), radiation alone ($n = 2$), or tamoxifen therapy ($n = 3$). Five were alive with stable disease and normal function status until the end of follow-up, and 3 had uncontrollable progressive disease, including 1 with mesenteric disease and Gardener's syndrome who died after 3 operations.

Intra-abdominal desmoid tumors

Thirteen patients who had intra-abdominal disease were treated, of whom 3 were complicated with Gardener's syndrome. Among these patients, 7 had complete resection with negative margins, 1 had R1 resection and subsequent tamoxifen therapy, and 5 had R2 resection and adjuvant radiotherapy. The 5-year EFS rate of this subgroup was 69.2%. Information on these patients is detailed in Table 2.

Table 2 Clinical parameters and treatment of intra-abdominal tumors

Patient	Gender/age (y)	Location	Diameter (cm)	Treatment	Outcome
1	F/36	Right upper quadrant of abdomen (mesentery)	15.0	R0	DFS
2	F/27	Right quadrant of abdomen (mesentery)	5.0	R0	DFS
3	F/33	Whole abdomen and pelvic cavity	35.0	R2, radiation, tamoxifen	SD
4	F/50	Pelvic cavity (between the bladder and rectum)	12	R1	DFS
5	F/26	Right ischiorectal fossa	11.0	R2, radiation	PD (relapse)
6	M/71	Umbilical region (mesentery origin)	4.0	R0	DFS
7	F/21	Left lumbar region	12.0	R2, radiation	PD (relapse)
8	F/30	Left iliac fossa next to wall of sigmoid colon	12.0	R0	DFS
9	M/34	Umbilical region (mesentery origin)	13.0	R2, tamoxifen	SD
10	F/40	Multifocal disease (abdominal cavity and mesentery root)	9.0, 12.0	R2, radiation	PD (relapse)
11	M/45	Mesentery border	3.0	R0	DFS
12	M/42	Umbilical region (mesentery origin)	14.0	R0	DFS
13	M/34	Left iliac fossa	6.0	R0	DFS

DFS = disease-free survival; F = female; M = male; PD = progressed disease; SD = stable disease.

Watchful waiting

Nine patients who underwent biopsy alone in group III and 11 patients with recurrences who received observation alone were regarded as the watchful waiting group. The median follow-up duration was 45 months (range, 24 to 90 months). Among these patients, 4 presented with progression during follow-up. The median delay to progression was 15.3 months (range, 7.8 to 41 months). The estimated 5-year EFS rate was 71.2%. Fourteen patients had stable disease. Spontaneous regressions occurred in 2 patients. None of these patients presented with progression at the end of follow-up.

Comments

Desmoid tumors are rare diseases, with intrinsic benign features but an infiltrative growth pattern, which can be derived from various parts of the body. The infiltrative growth of the disease across the viscera and into the muscles, bone, and neurovascular structure reduces the likelihood of complete excision and enhances the local recurrence rate. Randomized studies or even large prospective studies on which to base treatment strategies are currently lacking. Additionally, the heterogeneity and highly unpredictable natural course of the disease have added to the difficulty of standardizing treatment.

For resectable disease, surgery has historically been considered by many institutions to be the key to eradication.^{6,8,9,11,17-20} Limb-sparing and function-preserving surgery has been accepted as a mainstay of surgical treatment. Because of its benign characteristics and the possibility of spontaneous regression, a conservative, watchful waiting approach for treating stable disease has been advocated for selected patients in the literature.^{8,10} Therefore, for this disease, the role of surgery and suitable surgical patients need reevaluation and further focus. We retrospectively analyzed 22 years of DF treatment history at our institution, the largest series so far reported in China, aiming to evaluate the treatment of this disease, identify independent prognostic factors, and find suitable surgical candidates.

The prognostic significance of positive margin remains controversial.⁶⁻⁹ Surgical margin status has been shown to correlate with risk for recurrence in our previous results and other published studies.^{7,21-23} However, some major studies have found no correlation.^{8,9,24} This series included patients beyond the scope of previous studies including those with intra-abdominal or macroscopically resected disease and patients undergoing only biopsy. Margin status had a significant impact on EFS in univariate analysis but was not an independent predictor in a multivariate model. Patients who underwent R0 resection had the best outcomes, whereas those who underwent R1 resection had similar outcomes to those with R2 resection or biopsy alone. Therefore, we concluded that patients benefited more from

surgery with margin-negative resection. However, no functional impairment or cosmetic defect was warranted to achieve surgery with negative margins, because it would not independently influence EFS. Conversely, if R0 resection is not deemed feasible, nonsurgical treatment is preferable.

Patients in group II had similar EFS rates to those in group III, which is at variance with other series. Uncontrolled differences between the patient groups, differences in adjuvant treatment over the 20 years of this retrospective study, and varying individual tumor characteristics may be the confounding factors. Furthermore, 9 unresectable patients treated with biopsy and watchful waiting were included in group III, which may have also caused variance. If these patients are removed from group III, its 5-year EFS rate declines to 51.2%, which is lower than that of group II but not significantly. The EFS of completely resected patients (R0 or R1, groups I and II) in our study was higher than reported in other studies, which may be mostly attributable to the high R0 resection rate (66.3% of surgeries) and high percentage (42.5%) of abdominal patients, who tend to have better outcomes than those with tumors in other sites. The ratio of male to female patients in our series was 1:3.4, which was greater than that in other reports (range, 1:1.8 to 1:3.7).^{8,9,21,22,24} In our previous study, the ratio reached statistical significance by log-rank tests ($P = .025$) but was not significant in multivariate analysis. In this study, the percentage of female patients was also high, reaching 94.5% in the abdominal group and 64.2% in the extra-abdominal group. If included in the multivariate regression model, patient gender would not change the significance of tumor size and admission status.

Prognostic factors for EFS vary, even among studies at large institutions.^{6-9,21,22,24} In our study, tumor size and admission status were identified as independent prognostic factors. Patients with tumors >10 cm, recurrent disease, tumors in the extremities or head and neck area, or macroscopically positive resection had worse outcomes. Bonvalot et al²¹ also identified microscopic margin status and tumor site as significant prognostic factors and found that extremity tumors had a worse prognosis. Tumor size was also confirmed to be a prognostic factor by Gronchi et al⁸ and Lev et al.⁹ Fiore et al¹⁰ reported that no significant clinical variables emerged as independent predictors of outcomes for patients treated with conservative therapy, but there was a trend toward worse outcomes for tumors in the trunk and those >10 cm. Even allowing for the heterogeneity of the disease and varying treatments in the reported literature, inherent characteristics such as tumor size and site have been identified in this and multiple previous studies as independent prognostic factors, regardless of whether resection was accomplished with positive or negative margins.

Although tumor site has not been identified as an independent prognostic factor in the literature, outcomes varied among different locations, which might be ascribed

to the adjacency to vital structures and difficulty of resection. Abdominal wall DF tends to have fairly better outcomes, with reported 5-year EFS rates of 61.7% to 88%. Extra-abdominal tumors, including head and neck–originating tumors, have aggressive behavior and usually invade to adjacent bone or vital structures, causing difficulty of complete resection and a high propensity for recurrence. The outcomes of head and neck tumors are rarely reported in large samples and also vary by different tumor sizes and resection status in the literature. Salas et al²⁴ reported 28 of 426 patients with head and neck tumors in their series. The 5-year progression-free survival rate was 60.4%, which was comparable with the 69.2% rate in our group. In our study, the mean size of head and neck tumors was 7.4 cm, compared with 5.2 cm in patients with chest and abdominal wall disease. Nineteen of 36 patients (52.8%) underwent complete resection, compared with 85 of 91 patients (93.4%) in the chest and abdominal subgroup. The larger tumors and lower complete resection rate might lead to the inferior EFS in the head and neck subgroup compared with those in the chest and abdominal wall group.

For unresectable DF, the most effective approach is still unclear. Radiation, chemotherapy, or other systemic therapies (tamoxifen, sulindac, or nonsteroidal anti-inflammatory drugs) may have some efficacy in controlling progression.^{15,25–27} The efficacy of radiotherapy is supported by substantial evidence.^{17,28–30} Its use in unresectable disease has been widely reported in large referral centers, usually with adequate long-term palliative control. In addition, adjuvant radiation has also been shown to be effective for local control and to offset the adverse effect of positive margins.^{28–30} However, univariate analysis in our series failed to show any positive effect for adjuvant radiation therapy. Radiotherapy was administered to 10% of R0 patients and to 57% of R1 and R2 patients. The groups were not homogeneous, which is a confounding factor in comparing the impact of radiotherapy retrospectively. In subgroup analysis, patients who received radiation in groups II and III had relatively better outcomes than those did not, but this was not found in group I. Some high-risk patients with complete resection and adjuvant radiation had subsequent recurrent disease, which may have eliminated the effectiveness of radiation in univariate analysis. Nonetheless, the efficacy of radiation for palliative control of unresectable or margin-positive resected disease is still evident on the basis of our results.

Pharmacologic agents are usually the initial agents of choice when managing abdominal-type desmoid tumors or patients complicated with familial adenomatous polyposis. Hansmann et al²⁵ reported their successful experience with high doses of tamoxifen and sulindac as a first-line treatment for familial adenomatous polyposis–associated tumors, achieving a 90% local control rate, which was <40% in sporadic DF. De Camargo et al³¹ concluded that systemic therapy can be successful in patients with DF. Antiestrogens and anthracycline-containing regimens appear to be associated with a higher radiologic response

rate. In our series, we administered tamoxifen at 30 mg/day in 34 selected patients with tumors >15 cm in diameter. A local control rate of 64.7% was demonstrated. However, further case-control study is needed to evaluate the efficacy of systemic therapy.

A conservative, watchful waiting approach has been recently advocated for recurrent, locally advanced, and even primary disease.^{10,21,24} Outcomes with watchful waiting have been reported as equivalent to R0 resection or medical treatment. Fiore et al¹⁰ reported that in 142 patients with DF who received either observation or medical treatment, 71 (50%) were free of progression and 3 had spontaneous regression after 5 years of follow-up. In our watchful waiting group, 4 patients (20%) developed progressive disease, 14 had stable disease, and 2 had spontaneous tumor regression until the end of follow-up. The median delay to progression in these patients was similar to that of the whole group (15.3 vs 16 months), and the 5-year EFS rate was comparable with that of patients treated with surgery (71.2% vs 76.2%). Thus, considering the unique biology of DF, conservative treatment strategies merit further investigation in unresectable primary diseases and even in resectable diseases.

Conclusions

Our study included the largest series of desmoid tumors in China. Inherent characteristics such as tumor size and a history of recurrence have been identified as independent prognostic factors, which is consistent with other reports in the literature. Surgery is warranted if it can be R0 and function sparing. Nonsurgical modalities or watchful waiting may be a better choice for unresectable disease. However, patients in our study were intentionally treated with surgical approaches, and no specific nonsurgical treatments were intended. The optimal criteria for choosing surgery or nonsurgical treatments still need further controlled study.

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